**Introduction**
This report outlines the design and implementation of a bandpass filter bank and envelope extraction to process sound signals for cochlear implants. It details the steps and considerations involved in the filter design, including frequency range distribution, filter type selection, and envelope extraction method.

For this implementation, a recording of a clarinet sound is used as the input signal. The results and plots presented in this report correspond to the processing of this audio sample.

**Generating Filter Bands**
Cochlear implants typically use 12 to 22 channels to balance auditory resolution with processing efficiency [1]. For this design, the average of this range, 17, is chosen for the number of frequency bands (N=17).

According to Michael Filimowicz, a lecturer at Simon Fraser University, the human auditory system processes sound intensity and frequency on a non-linear, logarithmic scale [2]. To reflect this, the 17 frequency bands in this design are spaced logarithmically. This scale ensures that more bands are allocated for lower frequencies, given that the human ear perceives more sounds at a lower range. For instance, speech and environmental sounds inherently exist at lower frequencies, and variations in pitch are also more closely spaced at lower frequencies [3]. In contrast, higher frequencies are spaced more widely [4]. The distribution of the filter bands therefore mimics the human auditory system's response to sound, making the design more effective for cochlear implant simulation.

**Filter Selection**
In this cochlear implant filter design, the simulation of natural hearing is prioritized. The Butterworth filter is selected for its smooth and consistent frequency response that minimizes distortion within each band. While alternative options like Chebyshev and elliptic filters offer sharper roll-offs, they introduce ripple and phase distortions that could compromise signal clarity. In contrast, the Butterworth filter's gentle roll-off from the cutoff frequency ensures that each frequency band is defined without abrupt cutoffs, preserving clean and undistorted signals [5].

**Butterworth Filter**
Unlike Chebyshev or elliptic filters, which sacrifice passband flatness for sharper roll-offs, the Butterworth filter provides a smoother, gradual transition from the passband to the stopband [5]. It provides a uniform response without oscillations, achieving a maximally flat passband [6]. In combination with its phase linearity characteristics, the Butterworth filter ensures minimal signal distortion and maintains a consistent frequency response across the passband [5][7].

The flat passband ensures that all frequencies within the band are treated uniformly, avoiding amplitude distortions, while the minimal phase distortion helps maintain the temporal structure of the signal [7]. These properties are especially important in cochlear implant applications, where preserving the spectral and temporal characteristics of the signal is critical for accurate auditory perception.

In the actual filter design, a low-order Butterworth filter (order 2) is chosen to attain both a gradual roll-off and computational efficiency to prevent phase distortion [5][8]. Although higher-order filters offer sharper and more accurate cutoffs, they could introduce excessive passband ripple and computational complexity [8]. Thus, by keeping the order low, these complications are minimized.

**Bandpass Filtering**

The Butterworth filter is applied as a bandpass filter to isolate the frequency bands within the audio signal. Each bandpass filter removes frequencies outside its range while ensuring that the retained frequencies remain clean and undistorted. This approach helps maintain a smooth and accurate representation of the original signal, which is essential for simulating the natural range of human hearing in a way that sounds authentic to the input.

Specifically, each band is processed using a second-order Butterworth bandpass filter, implemented in the `bandpassFilter` function. The cutoff frequencies for the filter are normalized as**:**

$$Wn = \frac{Band}{Sampling\ Frequency/2}$$

Here, the sampling frequency (*fs*) is obtained from the input audio file and represents the number of samples per second in the audio signal [9]. This value is critical because it determines the Nyquist frequency, (*fs* / 2), which is the maximum frequency that can be represented in the signal [9]. The normalized frequency range, *Wn*, is a two-element vector specifying the lower and upper bounds of the frequency band, relative to the Nyquist frequency. This ensures compatibility with MATLAB's filter functions, which operate in the range [0, 1], where 1 corresponds to the Nyquist frequency [10].

```
Wn = band / (fs / 2);
```

The filter design is performed using MATLAB's built-in `butter` function, which calculates the coefficients for a second-order Butterworth filter [10]. The filtering process itself is carried out using MATLAB's `filter` function, which applies the bandpass filter to the input signal, isolating the specific frequency components within each band [16].

```
[b, a] = butter(2, Wn, 'bandpass');
bandOutput = filter(b, a, inputSignal);
```

The following figures show the output signals of the lowest and highest frequency channels, respectively. The output of each bandpass filter represents the signal energy within its corresponding frequency range and is then used for envelope extraction in the next step.

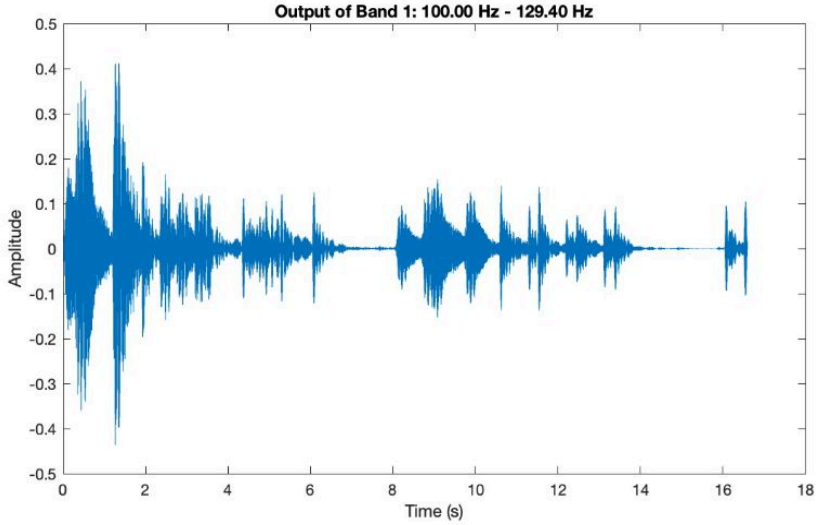



*Figure 1: Output Signal of the Lowest Frequency Channel*

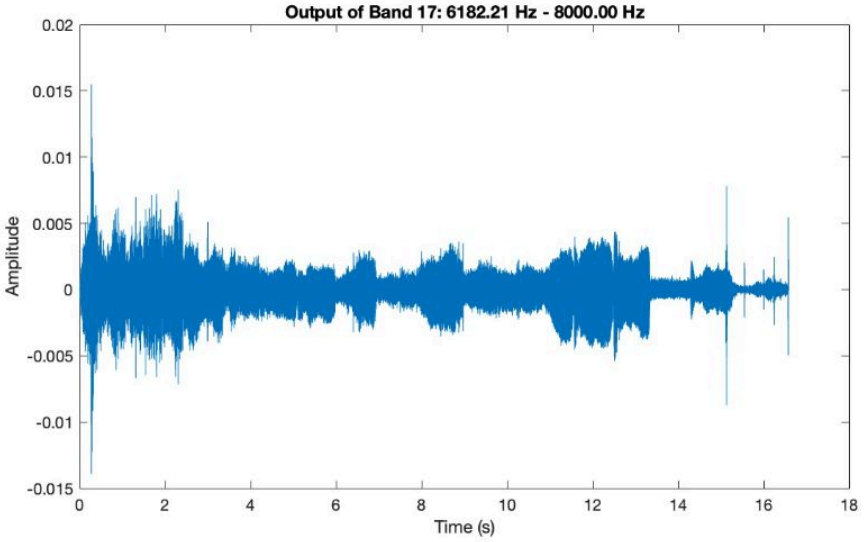



*Figure 2: Output Signal of the Highest Frequency Channel*

**Envelope Extraction**

Envelope extraction of a signal involves isolating its amplitude variations, which can be processed with a rectifier and low-pass filter. Input signals from sounds are rapidly oscillating waveforms, so the envelope outlines the general trend, or outer boundary, of a signal's amplitude over time [11].

The first step of envelope extraction is rectification—a signal processing technique in which the signal is converted to a unipolar signal [12]. This means that the signal is processed so that all signal amplitudes are positive, which removes the rapid oscillations about zero while retaining the envelope information [12]. This can be computed in MATLAB using the `abs` function:

```
rectifiedSignal = abs(bandOutput);
```

The rectified signal is then processed with a second-order Butterworth low-pass filter, which allows the low-frequency components of the filter to pass while suppressing the high-frequency components. The high-frequency components represent the rapid changes from the sharp transitions of the waveform. The low-frequency components represent gradual trends, therefore the low-pass filter effectively isolates and smoothens the signal to extract its envelope [13].

A Butterworth filter is selected for its maximally flat frequency response within the passband, which minimizes signal distortion, as previously mentioned. This ensures that all low-frequency components of the signal are preserved without distortion, which is especially important for the cochlear implant. The Butterworth filter also provides a smooth transition from passband to stopband, which is applicable for this envelope extraction, where the abrupt variations and sharper cutoffs are avoided [14] [15].

Envelope extraction is computed using the MATLAB `butter` function [10] [16]:

```
Wn_lowpass = lowpassCutoff / (fs / 2);
[b_lowpass, a_lowpass] = butter(2, Wn_lowpass, 'low');
envelope = filter(b_lowpass, a_lowpass, rectifiedSignal);
```

The following figures present the results of envelope extraction performed on the clarinet audio input signal for the lowest and highest frequency bands.

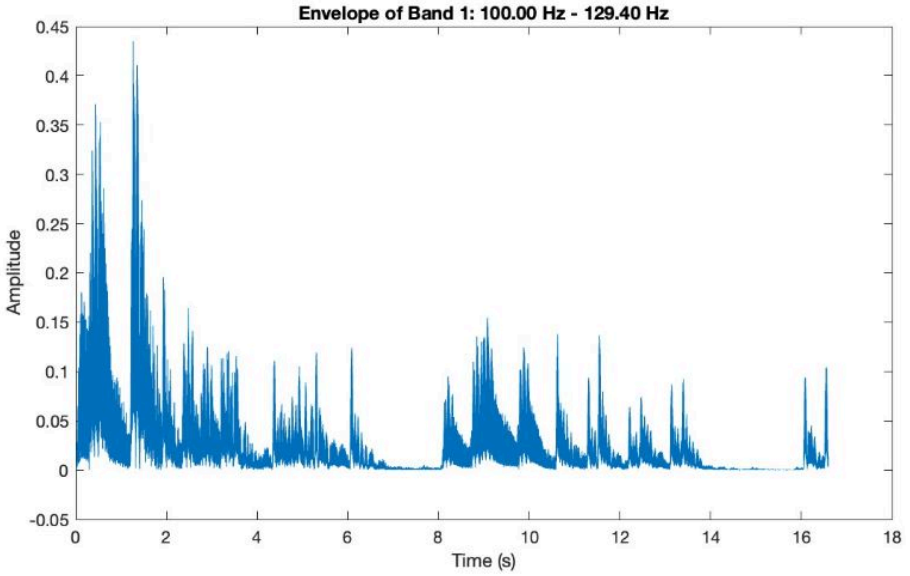



*Figure 3: Extracted Envelope of the Lowest Frequency Channel*

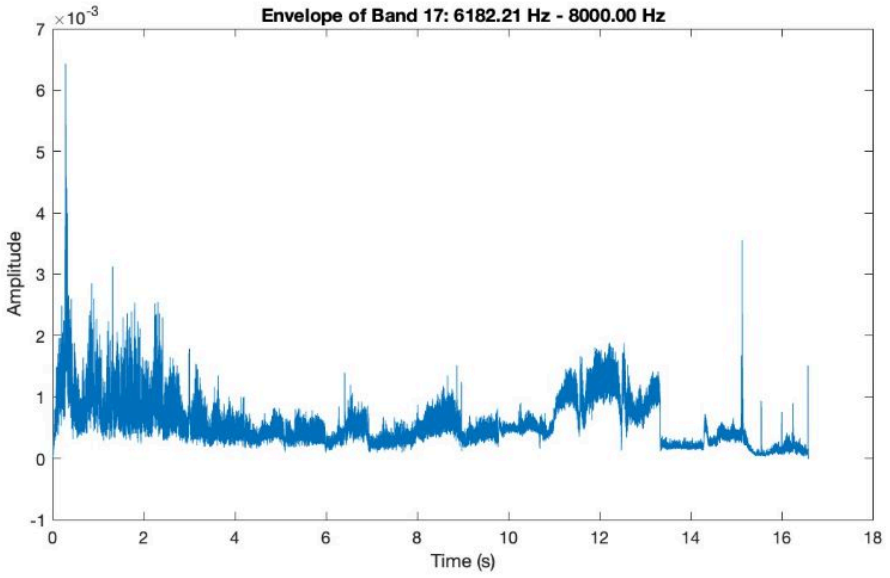



*Figure 4: Extracted Envelope of the Highest Frequency Channel*

**Conclusion**

In conclusion, to design an effective cochlear implant filtering system, the preservation of spectral and temporal characteristics of input signals is prioritized. This is achieved by selecting a bank of 17 logarithmically spaced frequency bands, which are then filtered using a Butterworth filter. The corresponding output signals are then rectified and passed through a Butterworth low-pass filter to extract their envelopes.

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